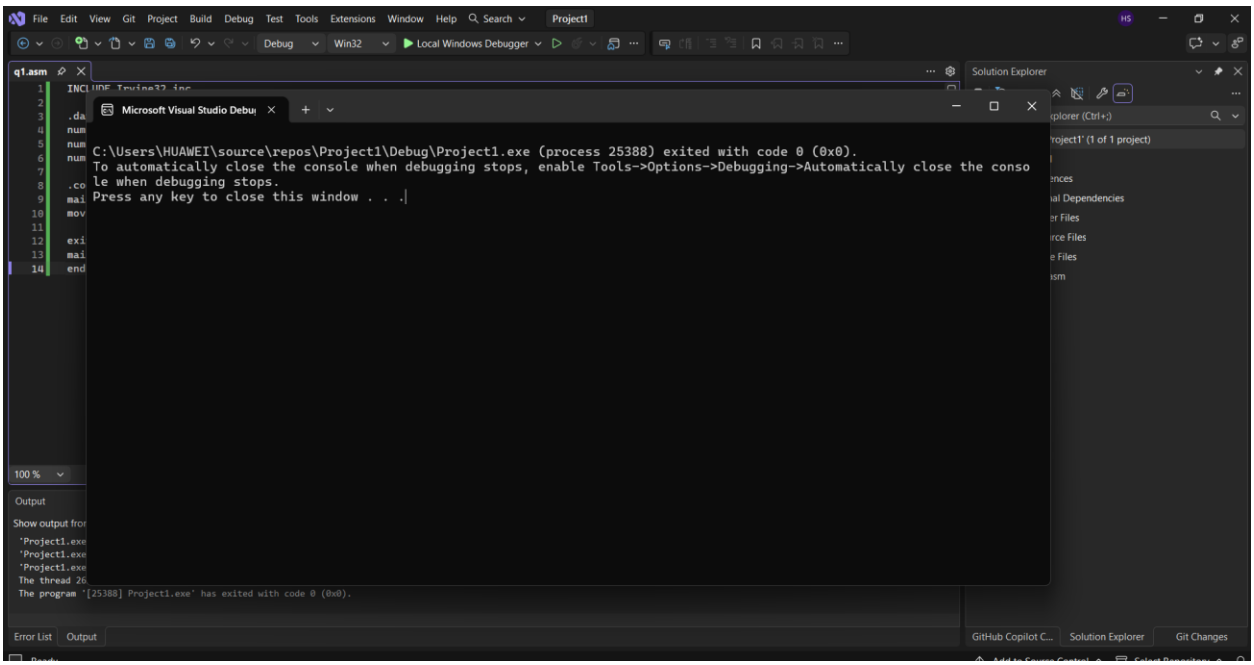
Q1)

```
INCLUDE Irvine32.inc

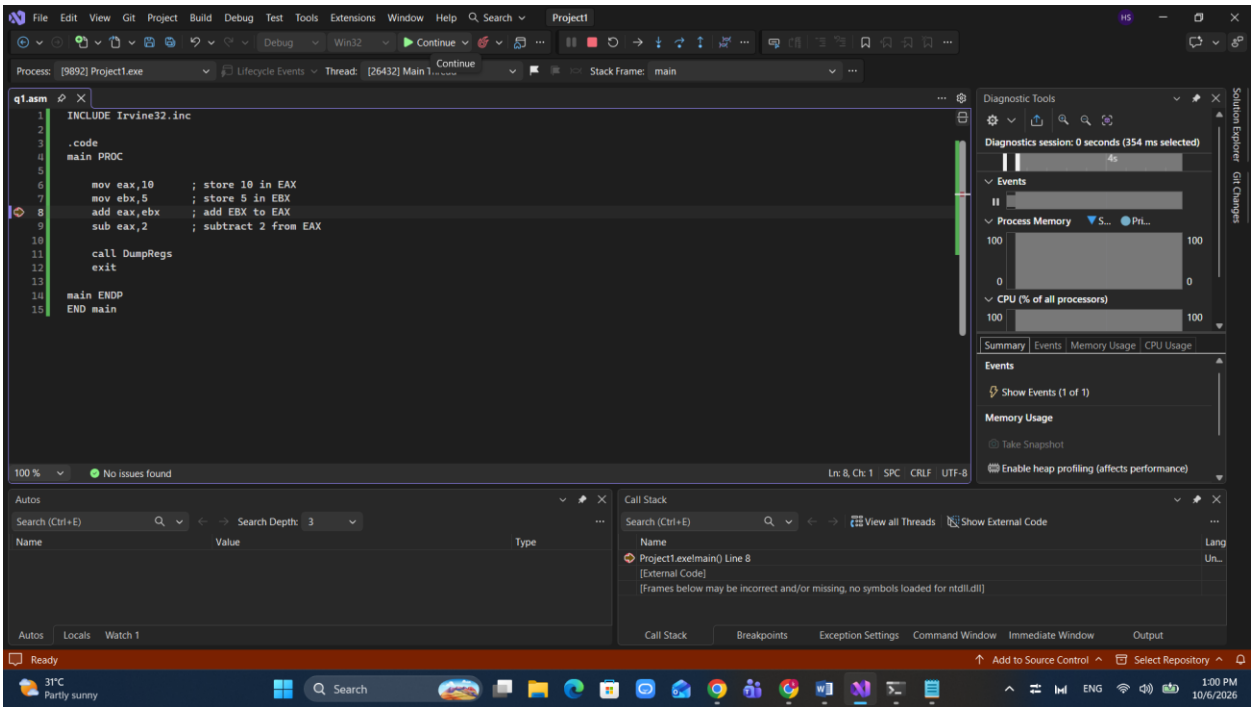
.data
num1 byte 25
num2 word 300
num3 dword 5000

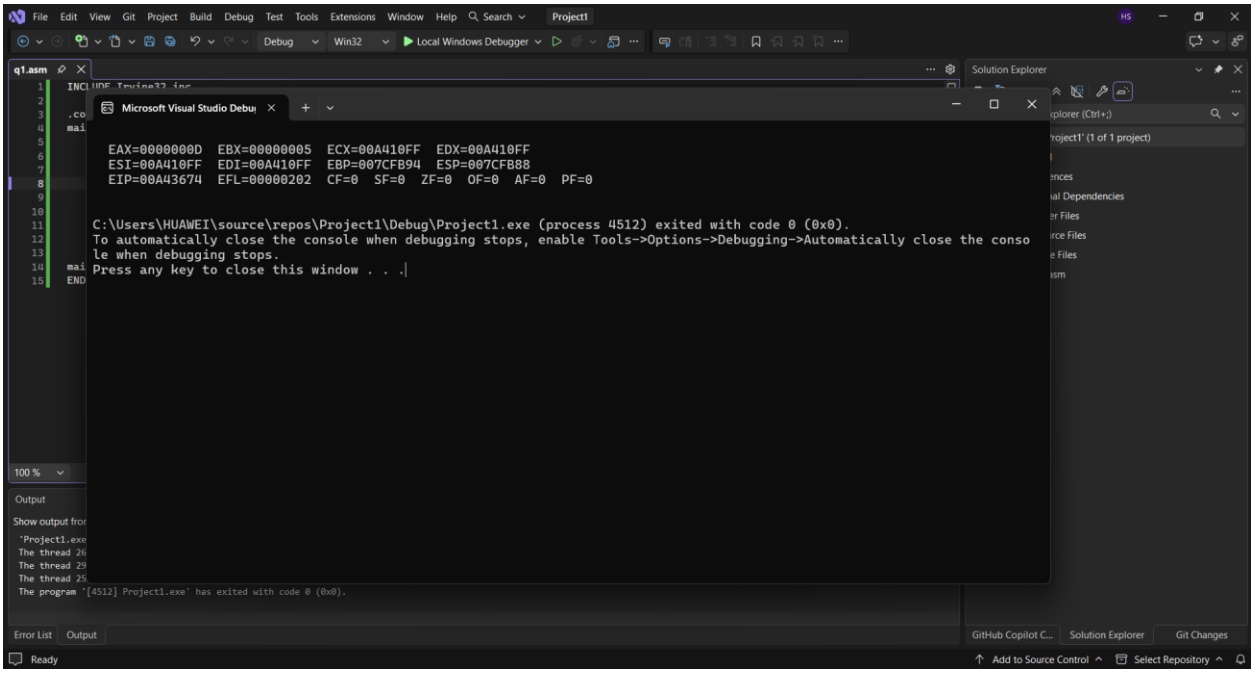
.code
main proc
mov eax,0

exit
main endp
end main
```



Q2)






```
INCLUDE Irvine32.inc

.code
main PROC

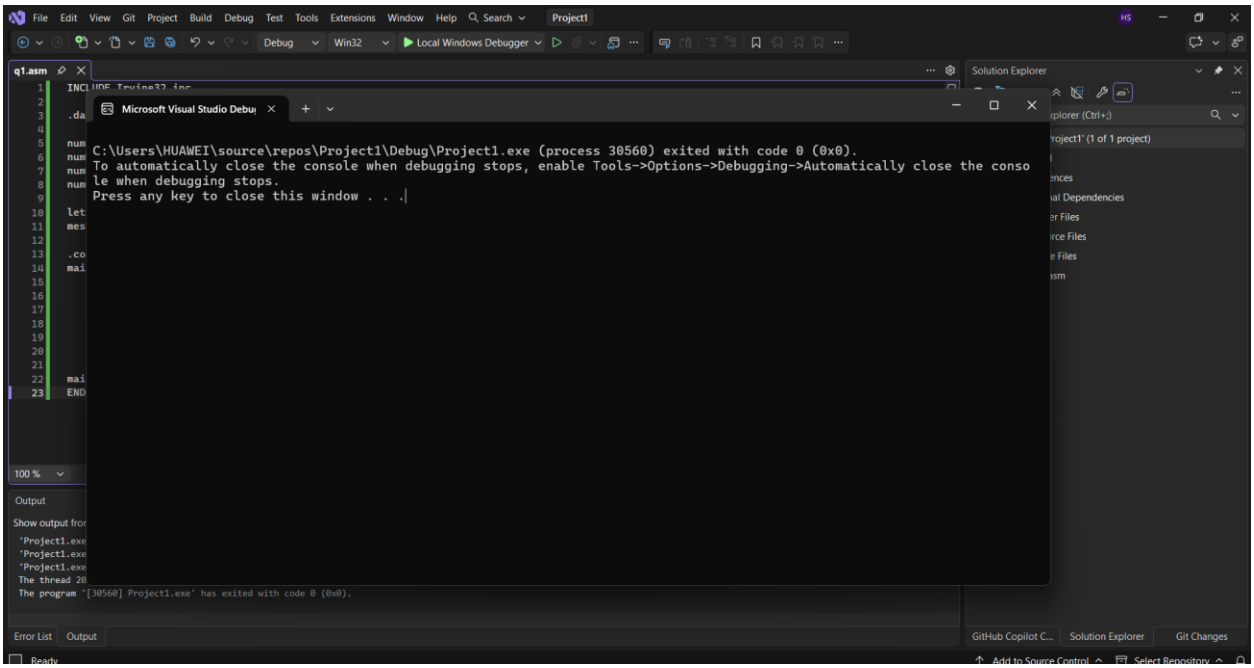
    mov eax,10      ; store 10 in EAX
```

```
    mov ebx,5       ; store 5 in EBX
    add eax,ebx     ; add EBX to EAX
    sub eax,2       ; subtract 2 from EAX

    call DumpRegs
    exit

main ENDP
END main
```

Q3)




```
INCLUDE Irvine32.inc

.data

numByte DB 25           ; 8-bit integer
numWord DW 300          ; 16-bit integer
numDword DD 5000        ; 32-bit integer
numQword DQ 100000      ; 64-bit integer

letter DB 'A'           ; single character
message DB "COAL Lab Task 3",0   ; string

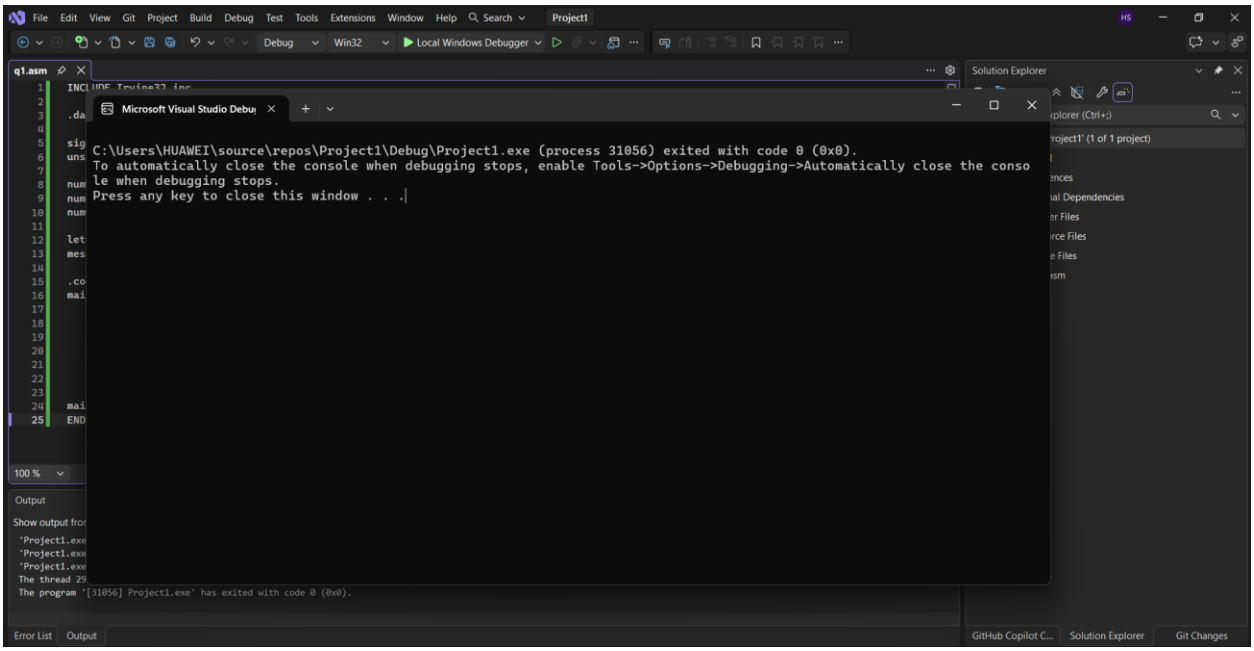
.code
main PROC

    mov eax,0           ; simple instruction for debugging
    mov ebx,1           ; another instruction
    add eax,ebx         ; EAX becomes 1
```

```
    exit

main ENDP
END main
```

Q4)




```
INCLUDE Irvine32.inc

.data

signedByte SBYTE -50
unsignedByte BYTE 200

numWord WORD 30000
numDword DWORD 500000
numQword QWORD 1000000

letter BYTE 'A'
message BYTE "COAL Lab Task 4",0

.code
main PROC

    mov eax,0
    mov ebx,1
    add eax,ebx

    exit
```

```
main ENDP
END main
```